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Liana Abundance Patterns: The Role of Ecological Filters during Development

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ABSTRACT

We studied three different size classes of liana abundance representing proxies for three different life stages and aimed to identify the sequence of ecological filters that have led to current patterns of liana abundance. We tested the relationship between vegetation structure (including antagonistic support types) and soil texture on liana abundance, using 40 plots (1 and 0.25 ha) set at least 1 km apart, and distributed over 64 km^2 in a Central Amazonian terra firme forest. Three support types were considered: palms, thin trees and an index of vegetation structure. Liana size classes responded hierarchically to ecological filters: larger size classes were progressively less associated with the environmental variables, while different aspects of vegetation structure were related to individual size classes. This hierarchical pattern suggests that selection mechanisms change throughout liana life cycles. Our results show that vegetation structure is an important predictor of liana abundance at a mesoscale and highlights the importance of considering the spatial structure in studies of tropical liana communities.

Key words: Amazonia; Brazil; palms; soil characteristics; spatial dependence; terra firme; vegetation structure.

DESPITE MANY STUDIES ON LIANA DISTRIBUTION PATTERNS IN RELATION TO ENVIRONMENTAL PREDICTORS (annual rainfall and seasonality; Schnitzer 2005, DeWalt *et al*. 2010; altitude: Parthasarathy *et al*. 2004; soil structure and chemical composition: Laurance *et al*. 2001, Ibarra-Manríquez & Martinez-Ramos 2002, Malizia *et al*. 2010; canopy height: DeWalt *et al*. 2000, Parthasarathy *et al*. 2004; disturbance intensity: Ibarra-Manríquez & Martinez-Ramos 2002, Rice *et al*. 2004, Malizia *et al*. 2010; and edge effects: Laurance *et al*. 2001), none of these studies has taken into account the potential impact of the variation of the developmental stages of lianas to the overall abundance of lianas in individual communities.

It is clear, however, that successful plant regeneration depends on being able to overcome the selective filters throughout the sequential stages of development (Wang & Smith 2002, Jordano *et al*. 2004). More specifically, the relationship between a plant and its environment is expected to vary according to changes in the environment and changes in a plant's individual needs. Therefore, studies that aim to determine the factors shaping the structure of a community need to take into account the variation in plant sizes. For example, soil is known to be particularly important in determining seed germination and seedling survival and young individuals are expected to be more affected by soil properties than older ones, while at older life history stages liana growth is more closely associated with supporting tree structure (Kurzel *et al*. 2006).

The abundance of lianas is dependent on tree structure, which affects liana's ability to access the canopy (Putz & Holbrook 1991). Most Neotropical lianas have tendrils or are stem-twiners enabling them to attach and climb from the lowest forest strata to the canopy, but their ability to do so is constrained by understory conditions. Lianas' abilities to climb into the canopy is facilitated if the branches or stems of understory trees are between 0.5 and 10 cm diam (Putz & Holbrook 1991, Balfour & Bond 1993). Palms can displace or damage lianas by growth of new palm leaves which can cut liana stems (Vleut & Pérez-Salicrup 2005), by liana seedlings being buried under fallen palm leaves or destroyed by increasing abundance of insects and pathogens under palm leaf litter (Putz 1980, Denslow *et al*. 1991, Harms *et al*. 2004), and by attenuation of light reaching the understory (Chazdon 1986). Because understory palms are abundant in Central Amazonia, we expect forest areas with high understory palm density to contain fewer trees and lianas in the lowest forest strata.

As lianas establish sufficiently to begin climbing, we expect tree structure and availability of light to limit liana abundance in the canopy. Forest structure is correlated to the light penetration (Wright & Van Schaik 1994, Drever & Lertzman 2003), although such relationships are complex (Montgomery & Chazdon 2001). Therefore, any representation of forest structure will inevitably also, to a degree, represent light availability.

Understanding the process that shape liana abundance patterns therefore requires an evaluation of different liana life-history stages, and so we address the relationship between the distribution of three size classes of lianas and selected ecological filters in 64 km^2 of undisturbed Amazon forest. We explore the relationships between the abundance of the liana size classes and four independent variables: soil texture, palm density, density of thin trees ($1.0\,cm \geq dbh < 10.0\,cm$), and an index of vegetation structure (IVS) that represents the distribution of tree sizes over 10 cm dbh. Owing to possible spatial autocorrelation, which might generate spurious correlations among variables, we additionally included spatial location (space) as an independent variable.

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METHODS

STUDY AREA.—This study was conducted at Ducke Reserve (Reserva Florestal Adolpho Ducke) of the Instituto Nacional de Pesquisas da Amazônia (INPA), located 26 km northwest of Manaus (02°55′ S, 59°59′ W at the reserve headquarters). Mean annual temperature is around 26°C and mean annual rainfall is around 2400 mm (SD = 400, between 1965 and 1980), with a dry season between July and October (Marques-Filho *et al.* 1981).

The reserve covers 10,000 ha (10 km × 10 km) of terra firme tropical rain forest. The vegetation is characterized by a closed canopy 30–37 m high (with emergent trees reaching 40–45 m), and an understory with abundant subterranean-stemmed palms, such as *Astrocaryum* spp. and *Attalea* spp. The soils represent a continuum from clayey latosols on the ridges, becoming sandier as the altitude decreases until the mineral fraction in the bottomlands is almost pure sand (Chauvel *et al.* 1987, Mertens 2004).

SAMPLING DESIGN AND LIANA DATA COLLECTION.—From the 72 permanent plots available at Ducke Reserve, we chose 40 plots. Initially, we categorized the original 72 permanent plots into five topographical categories according to their variation in terms of altitude and inclination. Subsequently, we used these five categories as basis to randomly select eight plots per category so that the broadest possible variation in topography was included. Hence, in total 40 plots were sampled over a 64 km^2 grid, formed by 8 km long trails (Fig. 1). All plots were at least 1 km from one another and from the edge of the reserve. More detail on the sampling design can be found in Magnusson *et al.* (2005) and Castilho *et al.* (2006). Each permanent plot was 250 × 40 m (1 ha total area) and was located lengthways along an altitudinal contour. The advantage of this design is that narrow plots following contour lines tend to enclose similar soil types as soil types vary with altitude in Central Amazonia (Chauvel *et al.* 1987, Luizão *et al.* 2004). Altitude is also related to soil water drainage, because it affects the depth to the water table (Hodnett *et al.* 1997a, b). All lianas with diam (D) ≥ 5 cm were sampled within these plots. Subsamples of 0.25 ha (10 × 250 m) were used to sample liana individuals with 1 cm ≤ D ≤ 4.9 cm. From August 2004 to October 2005 every individual liana stem rooted inside the plot and > 1 cm diam at 130 cm from the rooting point was tagged, counted and its diameter measured following Gerwing *et al.* (2006). For lianas with noncylindrical stems we measured the minimum section (S_{min}) and maximum section (S_{max}) and applied the equation $D = \sqrt{S_{min} \times S_{max}}$ to obtain a corrected estimate of stem diameter (Gerwing *et al.* 2006). The density of lianas in three diameter size classes (1–2.4, 2.5–4.9 and ≥ 5 cm) was also recorded. Since establishing developmental stages for different species is complex, we treated size classes as proxies for developmental stages. In general, lianas > 2.5 cm diam have already reached the canopy (Kurzel *et al.* 2006), but only when they reach the canopy do they expand in diameter (Putz 1995). We assumed that diameter size is positively associated to the height reached in the canopy during the process of climbing, as has been shown by Phillips *et al.*(2005). Therefore, the larger the liana diameter class, the higher is the assumed median height of liana leaves and greater is the portion of the forest canopy occupied (Phillips *et al.* 2005). Densities were standardized to a 1 ha area for analysis and the data sets used here are provided at the PPBio webpage, http://ppbio.inpa.gov.br/Eng

ECOLOGICAL FILTERS: VEGETATION STRUCTURE AND SOIL DATA.—Palms were sampled in a 4 × 250 m strip in each plot. Every

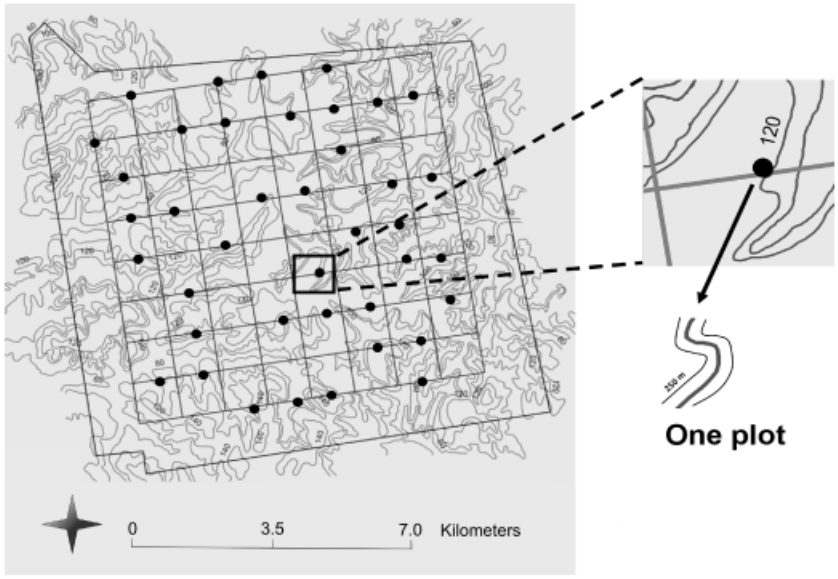


FIGURE 1. Trail system (straight lines) organized by east–west transect and position of sampled plots (filled circles) at Ducke Reserve, Manaus, Amazonas, Brazil. Altitudinal contour lines are shown for 20 m intervals, from 40 to 120 m asl. The detail shows an example of a plot following the altitudinal contour line.

individual rooted inside the plot and taller than 1 m height was counted (Costa *et al.* 2009). Most palms sampled were exclusive understory species (mean of 70 ± 17% SD), while most canopy species individuals (*e.g.*, *Oenocarpus bacaba* Mart. and *Oenocarpus bataua* Mart.) were juveniles growing in the understory.

The width of the plot sampled for trees varied according to size class. Individuals with dbh ≥ 30 cm were sampled in 1 ha, whereas subsamples of 0.5 and 0.1 ha were used for individuals with 10 cm ≤ dbh ≤ 29.9 cm and 1 cm ≤ dbh ≤ 9.9 cm, respectively (Castilho *et al.* 2006). Tree density was expressed as the number of individuals per hectare.

The overall density of trees was split into two variables: density of trees with 1 cm ≤ dbh ≤ 10 cm (thin trees); and density of trees > 10 cm dbh. 'Thin trees' represent stems in the ideal size for attachment of lianas with tendrils and stem-twiners (Putz & Holbrook 1991), and might represent an initial support, promoting the ascension of lianas from the understory after the self-supporting phase.

In order to represent the distribution of tree sizes in different forest strata above the understory, we created an IVS using the distribution of the density of trees larger than 10 cm dbh. A linear regression between seven tree-diameter size classes and the density within each class was run for each plot. Data were log-transformed before analysis to improve linearity of relationship between variables. The slope of the linear regression for each plot was used as the IVS. High values of IVS (steepest regression slopes) represent fewer large trees and a more open canopy. One assumption underlying the use of the IVS was that the difference in tree structure among plots is constant through time. Even if there is local change, due to gap openings (Condit *et al.* 1992), we expect that the tree structure of sampling units that far exceed the scale of gaps will tend to stay more similar within than among plots. This is a reasonable assumption for plots 250 m in length in the terra firme forests of Central Amazonia, which usually have small gaps (Almeida 1989). Although the complexity of tropical forest species composition and structure (allometry and crown size) prevent a simple relationship between the physical structure of the forest and light availability, the density of trees with 10–25 dbh has been associated with decreased light availability in the understory (Montgomery & Chazdon 2001). The slope of the IVS will be essentially the same among plots in situations where the proportion of small to large trees is similar, but their absolute values differ. In these cases, the variation in tree structure is not translated in different IVS values between plots, which is a limitation of the use of IVS.

Soil samples of 30 × 30 × 5 cm were taken at six points along the main axis of the plots, every 50 m, and bulked to produce a composite sample for each plot. Before analysis, samples were cleaned of roots, air-dried and sieved through a 2 mm sieve. Texture analyses were done in the Soil Laboratory of the Agronomy Department, at INPA (Mertens 2004).

In Ducke Reserve soil sand or clay contents are highly correlated with altitude ($r > 0.9$). In the same area, the gradient of altitude and soil particle size also describe the gradients of K^+, Mg^{2+}, Mn^{2+}, Na^+, Al^{3+} and $Al^{3+} + H^+$ (Castilho *et al.* 2006) and depth of the water table (Hodnett *et al.* 1997a, b). We therefore used sand content as a surrogate of these soil gradients: as sand content increases, nutrient availability decreases and water availability increases.

STATISTICAL ANALYSES.—Multicolinearity between independent variables was tested using simple correlations (Table S1). Density of trees with dbh < 10 cm were not used in the regression model due to the fact that the density of thin trees and palms are strongly correlated (collinearity), and hence cannot be used as this might lead to detrimental effects on the estimated regression parameters (Quinn & Keough 2002). Thus, linear multiple regressions were carried out to test if abundance of lianas in each of the three diameter size classes were related to soil sand content, palm density and IVS. A Mantel test was used to detect spatial dependence for each variable and spatial trends in data. We completed our analytical approach using partial Mantel statistics to check if relationships captured by multiple regressions with the original data would still hold when spatial auto-correlation was partialled out. Only variables that showed spatial dependence in simple Mantel tests were checked using partial Mantel. Significance of Mantel statistics was determined with 1000 permutations. Trend-surface maps and spatial correlograms (all-directional) were plotted to visualize the spatial structure of these variables, and we modelled the trend-surface by a polynomial of degree 2. Multiple regressions, Mantel tests between distance matrices and spatial correlograms were run with the Vegan and Pgirmess packages for the R environment (R Development Core Team 2006).

RESULTS

A total of 10,236 liana stems were sampled and used to estimate the density for the three diameter size classes. Lianas in the first diameter class (1 cm ≤ D < 2.5 cm) represented 69.3 percent of the stems per ha and their density was strongly correlated with the overall density ($r = 0.97$). Lianas > 5 cm of diameter represented only 7.6 percent of the total number of individuals per hectare. Nonetheless, this class represented half basal area (49.8%) and estimated biomass (A. Nogueira, unpubl. data).

Density of thin trees and palm density were correlated ($r = -0.6$, $P < 0.01$; Table S2); these variables were also spatially auto-correlated (Table 1), but partial Mantel statistics revealed that the correlation of thin tree and palm densities was real (Table 2). Because these variables have different hypothetical relationships with liana density they were retained in the conceptual model, although only palm density was used in the regressions due to the collinearity between variables (Table S1). Palm density was retained because this variable is expected to have both direct (*e.g.*, light reduction, increased physical damage) and indirect effects (through a reduction in the abundance of thin trees) on liana establishment and growth. The maps and spatial correlograms indicated that the spatial structure of the thin trees, palms and larger lianas had opposed distribution patterns from the western to eastern portions of the Ducke Reserve. Individuals of lianas > 5 cm of diameter were more abundant on the eastern portion, where thin trees were abundant and palm densities were not (Fig. 2).

TABLE 1. *Simple Mantel statistics (r) and the associated probabilities (P), for tests of spatial autocorrelation of all variables intended to be used in causal models. The Mantel test was considered significant after the Bonferroni correction of probabilities (0.05/7 = 0.0071), considering the 0.05 level of significance in seven tests.*

	Density of lianas $1 \geq D < 2.5$ cm	Density of lianas $2.5 \geq D < 5.0$ cm	Density of Lianas $D \geq 5.0$ cm	Palm density	Density of thin trees	IVS	Sand content
r	− 0.004	0.106	**0.243**	**0.174**	**0.192**	0.013	0.109
P	0.523	0.027	**0.001**	**0.007**	**0.001**	0.353	0.014

Bold numerals signifies significant tests.

The differences in IVS among plots represented the variation in the distribution of trees among size classes (Table S3). The extreme size classes (dbh > 70 cm and dbh between 10 and 19.9 cm) had a higher impact in the index values, but the correlation between extreme size classes and IVS were not high, suggesting that the index represents the whole distribution of tree sizes. The IVS was also correlated to thin (dbh between 1 and 9.9 cm) tree density ($r = 0.46$; $P = 0.003$).

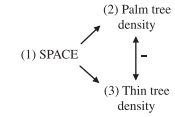
Environmental variables explained 47 percent of the variation in the overall density of lianas ($D \geq 1$ cm), which was negatively related to the soil sand content ($b_{std} = -0.27$, $P = 0.046$), palm density ($b_{std} = -0.36$, $P < 0.012$) and positively related to IVS ($b_{std} = 0.38$, $P < 0.005$). Each size class presented unique responses to the environment, with larger size classes being progressively less coupled with environmental variables (Table 3). Lianas in the first diameter class were associated with all predictors, while lianas in the second size class were not associated with soil sand content. Lianas in the third size class were not associated with either soil or IVS. The only effect that held across all three size classes was the negative effect of palm density, although both palm density and density of large lianas ($D \geq 5$ cm) were spatially auto-correlated (Table 3). The analysis of partial Mantel statistics indicated that the negative relationship between palm density and liana abundance was possibly an artifact of spatial autocorrelation (Table 4).

DISCUSSION

Successful plant regeneration depends on plants being able to overcome the selective filters along the processes of dispersion, germination, seedling survival and establishment (Jordano *et al.* 2004). Liana size classes differed in their responses to environmental factors. A hierarchical pattern of response was observed, with the smallest size class (thin lianas) being associated with all predictors, and the largest size class to none. This pattern suggests that environmental factors might act as ecological filters, selecting liana individuals during development. Soil sand content was a selective filter related only with the first liana diameter class, probably affecting plant germination and seedling establishment. The other factors, associated with structural components of the forest, acted as subsequent filters for lianas density. Therefore, some forest portions might favor liana's horizontal growth due to higher availability of supports and light, since lianas need the tree structure to reach and maintain themselves in the canopy, while portions with less or bad supports may restrict liana abundance. Similar to the pattern of abundance of lianas in the first size class, overall liana abundance was negatively related to soil sand content and palm density, and positively related to IVS. These results (and the strong correlation between the density of overall lianas and thin lianas) indicate that explanations of the overall liana abundance patterns will reflect mostly the factors influencing the abundance of small size classes, which are the most abundant lianas.

Even though several edaphic properties might represent good predictors of plant distribution, responses to these properties are variable (Sollins 1998). Other factors, such as movement of seed dispersers or predators (Wenny 2000), litter properties (Facelli & Pickett 1991, Vasconcelos & Luizão 2004), and herbivory (Fine *et al.* 2004, Boege & Dirzo 2005) may interact with the edaphic properties. In Central Amazonia, soil texture is associated with fertility and water drainage status of the soil (Chauvel *et al.* 1987, Luizão *et al.* 2004), both of which represent important factors for plant establishment and distribution (Sollins 1998, Clark *et al.* 1999, Clark 2002). Indeed, soil texture and altitude have been associated

TABLE 2. *Relationships among the variables space (1), palm density (2) and the density of small trees (3). The relationships were tested for each pair of variables, as stated in the first column. The last model (row) tests the relationship between palms and small trees, after considering the spatial effect. The last column illustrates the final relationships among variables in accordance with the computed results of the statistical tests.*

Prediction of the model if the model is true	Computed results	Statistical test used	Final relationships among variables
$1.2 \neq 0$	Yes	Simple Mantel	
$1.3 \neq 0$	Yes	Simple Mantel	
$2.3 \neq 0$	Yes	Pearson correlation ($r = -0.60$, $P < 0.001$)	
$(2.3).1 \neq 0$*	Yes	Partial Mantel ($r = 0.24$, $P = 0.015$)	

*The partial Mantel relation is denoted by (2.3).1 following Lengendre and Troussellier (1988).

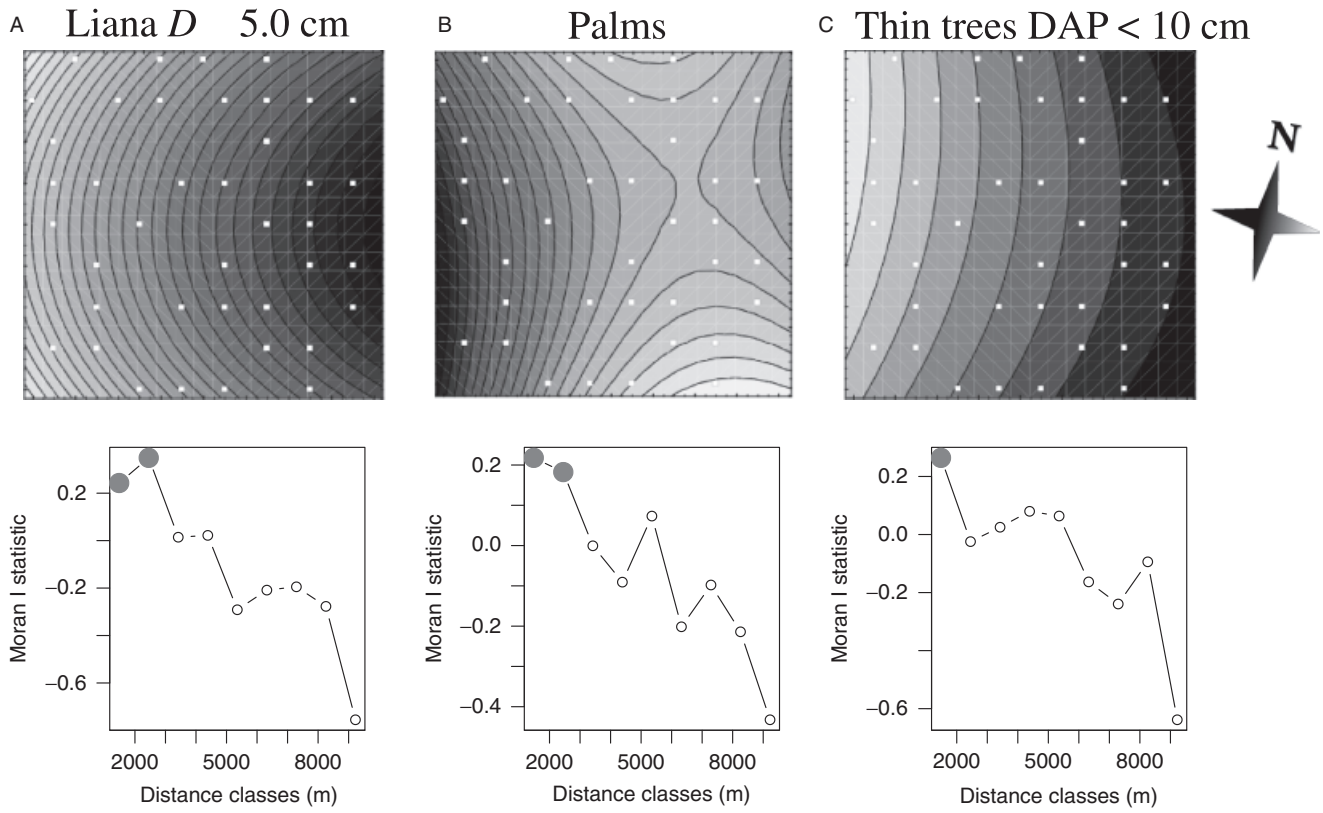


FIGURE 2. Trend-surface maps (above) and spatial correlograms (below) of the three variables positively related with space in simple Mantel test. Darker areas correspond to higher values of plant densities estimated by the second-order polynomial equation; and filled circles on the correlograms indicate the significant Moran I coefficients ($P < 0.05$).

with plant distribution of different taxonomic groups at the Ducke Reserve (Kinupp & Magnusson 2005, Costa *et al*. 2005, Costa 2006, Castilho *et al*. 2006, Drucker *et al*. 2008, Costa *et al*. 2009).

The areas at Ducke Reserve dominated by sandy soils are associated with bottomlands and are close to watercourses (Chauvel *et al*. 1987, Magnusson *et al*. 2005); therefore, these areas tend to be humid throughout the year. Lianas are known to have higher growth in seasonal environments or under water stress, compared with trees, due to deeper root systems and more efficient vascular systems (Schnitzer 2005). Under humid conditions, these advantages are expected to be less important, which would decrease the competitive ability of lianas over other plant groups; this might explain the lower abundance of lianas in sandy soils. Another possible explanation for the observed pattern is the fact that sandy soil generally has lower nutrient availability than clayey soils, and it could restrict the number of individuals and species of lianas that occupy such portions of the forest.

TABLE 3. *Standardized regression coefficients and associated probabilities (in parentheses) for each multiple regression between the density of lianas in three diameter size classes and the environmental factors soil sand content (%), palm density (individuals/ha) and IVS.*

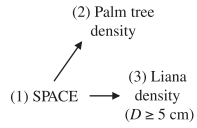
Density in each size class	Environmental factors			Overall results	
	Sand content	Palm density	IVS	R^2	P*
$1 \leq D \leq 2.4$ cm	− 0.30	-0.33	0.35	0.44	**0.000**
	(**0.03**)	(**0.02**)	(**0.01**)		
$2.5 \leq D \leq 4.9$ cm	− 0.17	− 0.33	0.37	0.36	**0.001**
	(0.25)	(**0.03**)	(**0.01**)		
$D \geq 5.0$ cm	− 0.05	− 0.36	0.23	0.24	**0.017**
	(0.73)	(**0.03**)	(0.13)		

*P is the probability associated with the complete model. Bold numerals signifies significant tests.

When lianas become larger, the forest physical structure and the availability of potential supports assume greater importance in the determination of abundance variation patterns. Because lianas are structural parasites (Stevens 1987), availability of satisfactory support represents the greatest restriction for height growth and maintenance on the different strata of the forest (Putz & Holbrook 1991). A low availability of supports may be seen as another ecological filter, restricting the local abundance of individuals within plots and generating patterns of variation in abundance at larger scales such as between plots at Ducke Reserve.

The abundance of lianas in all size classes was negatively related to the density of palms, corroborating our hypothesis that palms do not represent suitable supports for liana climbing and

TABLE 4. *Relationships for the variables space (1), palm density (2) and liana density with $D \geq 5$ cm (3). The relationships were tested for each pair of variables, as stated in the first column. The last model (row) tests the relationship between palms and small trees, after considering the spatial effect. The last column illustrates the final relationships among variables, in accordance with the computed results of the statistical tests.*

Prediction of the model if the model is true	Computed results	Statistical test used	Final relationships among variables
$1.2 \neq 0$	Yes	Simple Mantel	
$1.3 \neq 0$	Yes	Simple Mantel	
$2.3 \neq 0$	Yes	Pearson correlation ($r = -0.44$; $P = 0.005$)	
$(2.3).1 \neq 0$*	No	Partial Mantel ($r = 0.12$; $P = 0.067$)	

*The partial Mantel relation is denoted by (2.3).1 following Lengendre and Troussellier (1988).

could decrease the availability of light to other plants. Palm architecture may allow liana occupation (Vleut & Pérez-Salicrup 2005), but emerging leaves from the apical meristem disarrange lianas and may cut them. Frond shedding also leads to liana detachment (Putz 1980). Another possibility is that palms attenuate the light that reaches the understory, limiting liana seedling establishment. In a tropical forest of Costa Rica, light attenuation near forest ground was related to the abundance of short-statured palms, with crowns between 1 and 1.5 m (Chazdon 1986). Because most palms in our plots were in the understory, we believe that this filter selected lianas in the beginning of the climbing process in different ways.

Not only are lianas negatively associated with palms, tree abundance also decreased with palm abundance at Ducke Reserve. Seedlings below understory palms suffer more damage than in other microsites due to an increased activity of insects and pathogens (Denslow *et al.* 1991) and the fall of senescent leaves (*e.g.*, Peters *et al.* 2004). These damages reduce seedling survival, biomass and foliar area accumulation. Denslow *et al.* (1991) and Harms *et al.* (2004) reported an inverse relationship between palm and seedling density. Our results are in agreement with their findings as we also found lower liana and small-tree ($1 \text{ cm} \leq \text{dbh} < 10 \text{ cm}$) densities at sites with greater palm density. It is not possible from our results to establish whether palm density, thin trees density or both interfered directly on liana abundance. However, it is expected that the presence of thin trees, with stems in the adequate size for attachment of lianas with tendrils and stem-twiners would favor climbing and therefore increase the abundance of lianas. At Ducke Reserve palms probably affect directly liana survival, but we also must consider the indirect effect of reducing the density of trees that represent appropriate support to lianas. The spatial pattern of palms, thin trees and larger lianas was in accordance with this hypothesis, as lianas were less abundant on the western portion at Ducke Reserve, where palms are more abundant. On the other hand, lianas were more abundant on the eastern portion at Ducke Reserve, where higher abundance of thin trees is encountered.

We also hypothesized that lianas attaining the vertical limit of their understory supports (thin trees) should find a satisfactory neighboring support to continue climbing toward the largest light source in the canopy. Vegetation structure, as given by IVS and which represents both the availability of larger supports and light conditions, was also an ecological filter to the abundance of lianas. Ducke Reserve plots with steep slopes on IVS had higher density of lianas at the two first size classes. This result is compatible with the observation that liana density is higher in disturbed areas (DeWalt *et al.* 2000, Schnitzer *et al.* 2000, Laurance *et al.* 2001) and gap edges (Putz 1984), due to the higher light penetration and high tree density of intermediate-sized stems.

Balfour and Bond (1993) showed that canopy occupation by lianas can occur in two ways. At sites with tall canopies and lower availability of support, lianas occupied the canopy horizontally, coming from few trees that have already been climbed, leading to a low liana abundance. On the other hand, areas with greater availability of support, lianas had greater vertical growth, leading to large liana densities, as observed at Ducke Reserve. However, even areas with a favorable set of potential forms of support may not include large liana densities, due to constraints in the early stages of the climbing process (at Ducke Reserve represented by large densities of palms) and/or due to inadequate soil conditions. Campinarana sites at the Ducke Reserve, apparently have a favorable tree structure and good light penetration, and did not have high densities of lianas, possibly because the soil is sandy and understory palms are abundant.

It is possible that the absence of relationships between larger liana size classes and the environment, was only a reflection of past conditions (Mitchell 2005). Larger lianas have been in the forest for a longer period of time, and therefore may have been affected by conditions that no longer exist. Another possible source of uncertainty in the detection of relationships for this size class is associated with the sampling protocol used, because a smaller number of large individuals are generally included in the samples when small plots are used for all size classes. To prevent this problem we used plots with sizes proportional to the size class, therefore increasing the sampling effort so that the representation of different size classes was balanced. The coefficient of variation in abundance among plots was similar to all three size classes, suggesting that all size classes were well sampled.

CONCLUSIONS

Our results suggest that in 10,000 ha of Central Amazonian forest liana community structure is determined by factors that vary depending on liana size classes. Previous attempts to relate overall

liana abundance to environmental predictors have been limited to explaining the abundance of the most abundant size class—thin lianas—which might not be representative of large lianas that represent the majority of overall liana biomass.

Our results also suggest that vegetation structure and spatial variables should be incorporated into hypotheses seeking to explain patterns of liana distribution. Since lianas are structural parasites, the importance of supports has always been considered in analyses at local scales. Variables related to vegetation structure have, however, been overlooked or poorly explored at larger scales (Schnitzer 2005, van der Heijden & Phillips 2008). Here, we emphasize the importance of palms and alternative forms to represent tree size distribution (*e.g.*, IVS) to model the variation of forest structure relevant to lianas. In addition, the inclusion of a spatial component in our analyses reveled an unexpected spatial organization of different plant groups, also corroborating our initial hypothesis of the relationships between trees, palms and lianas.

At large scales, vegetation structure tends to covary with seasonality. Since less seasonal areas tend to have a higher number of palms (Salm *et al*. 2007), it becomes difficult to establish a causal relationship between these factors and liana abundance. The finding that a greater availability of favorable supports allows a greater abundance of lianas (or the inverse pattern with palms) is in accordance with the hypothesis that lianas grow more efficiently in seasonal areas, and may provide a link between local and global scale patterns.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article.

TABLE S1. *Product-moment correlation coefficients among independent variables on the 40 plots at Ducke Reserve*.

TABLE S2. *Basic statistics for the variables under study (40 plots sampled) at Ducke Reserve, central Amazonia*.

TABLE S3. *Index of Vegetation Structure (IVS) and tree density in three size classes for 40 1 ha permanent plots located at Ducke Reserve*.

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